

MATH 127 final quick review

No, this is not a replacement for actually reviewing slides and course notes. This might not even include everything because I might forget things. You should do practice exams and review your midterms, then do exercise problems from the course notes on topics that you are not comfortable with.

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Tip: For any problem on the final,

- Use **Table A** to choose the right general tool.
- Remember to state the hypotheses in **Table B** if a theorem/rule/method is involved.
- For approximations and error, consult **Table C**.
- For differentiation and integration, refer to the rules in **Table D** and the trig facts in **Tables E**.
- Curve sketching, see **Table F**.

Table A: What are you trying to find about $f(x)$?

Things about $f(x)$	Tools / ideas to try
Limit $\lim_{x \rightarrow a} f(x)$	Limit rules, simplification. Trig identities if it involves trig. Multiply by conjugate if it involves sum/difference of square roots. L'Hospital's rule, see Table B
Rate of change at $x = a$ Slope of tangent line at $x = a$ Derivative at $x = a$	Compute derivative using Table D .
Anti-derivative Integral $\int f(x)dx$ Solve differential equation $y' = f(x)$	Compute (indefinite) integral using Table D .
Area under curve Accumulated quantity Area between curves f and g Definite integral $\int_a^b f(x)dx$	Compute definite integral using Table D . For area between curves, integrate $f(x) - g(x)$.
Tangent line Linearization $L_a(x)$ Degree 1 Taylor polynomial $P_{1,a}(x)$	Point-slope form $L_a = f(a) + f'(a)(x - a)$
Degree 2 Taylor polynomial $P_{2,a}(x)$	$P_{2,a} = f(a) + f'(a)(x - a) + \frac{f''(a)}{2}(x - a)^2$
Critical points/numbers/values Inflection points Intervals of increase/decrease Intervals of concave up/down Local max/min	Refer to Table F .
Global max/min on a closed interval $[a, b]$	Use closed interval method in Table B .
Show that an equation $f(x) = k$ has a solution in $[a, b]$	Use IVT in Table B .
Approximate $f(x)$ near a point $x = a$	Use linear/quadratic approximation of Table C based on the question
Approximate expressions like $\sqrt[n]{1+x}$ or $(1+x)^n$ for x close to 0	Use binomial approximation of Table C

Table B: Theorem/Methods/Rules

Theorem	Hypotheses (MUST mention) + Conclusion
L'Hospital's rule	Hypotheses: The limit is of form $\frac{\infty}{\infty}$ or $\frac{0}{0}$, and Whatever result you get is a number or is $\pm\infty$. Conclusion: $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$.
Intermediate Value Theorem (IVT)	Hypotheses: f is continuous on $[a, b]$ 1) Check that $f(a)$ is below k and $f(b)$ is above k, 2) Conclude there exists c in $[a, b]$ such that $f(c) = k$.
Extreme Value Theorem (EVT)	Hypotheses: f is continuous on $[a, b]$ There f admits a global minimum and a global maximum of f on $[a, b]$. See below on how to find them.
Closed interval method	Hypotheses: f is continuous on $[a, b]$ 1) Find critical numbers of f in (a, b). 2) Evaluate f at all critical numbers in (a, b) and at endpoints a, b. 3.1) The largest value is the absolute maximum. 3.2) The smallest is the absolute minimum.
Fundamental Theorem of Calculus (FTC), Part I	Hypotheses: f is continuous on $[a, b]$ and $F(x) = \int_a^x f(t) dt$ Conclusion: $F'(x) = f(x)$.
Fundamental Theorem of Calculus (FTC), Part II	Hypotheses: f is continuous on $[a, b]$ and F is any antiderivative of f (there could be many, we just need any). Conclusion: $\int_a^b f(x) dx = F(b) - F(a)$
First derivative test	1) Find critical numbers c. 2) Split the x-axis using those c. 3) Compute the sign of f' on each piece by plugging in a number. 4.1) If f' changes $+$ $\rightarrow$ $-$: local max at c. 4.2) If f' changes $-$ $\rightarrow$ $+$: local min at c. 4.3) If no sign change: no local extreme at c.
Integration by substitution Reverse chain rule	Pattern: If the integrand includes some composition of function $f(g(x))$. 1) Set $u = g(x)$ and compute $du = g'(x) dx$. 2) Rewrite the integral completely in terms of u and du. 3) Integrate in u to get $F(u) + C$. 4) Substitute back $u = g(x)$ to write the answer in terms of x. 5) If it is a definite integral, you also calculate the modified bounds.
Classify discontinuity	Compare $\lim_{x \rightarrow a} f(x)$ from either sides: 1) If either is ∞, then vertical asymptotes/infinity discontinuity; 2) If two sides different, then jump discontinuity; 3) If two sides same but $f(a)$ doesn't exist, then removable discontinuity. 4) If two sides same and $f(a)$ exists, then continuous.

Table C: Approximation and Error Bounds

Tool	Formula	Conditions / error
Linear approx.	$f(x) \approx L_a(x)$ See Table A	Good when x is close to a with error $ \text{error} \leq \frac{M}{2} x - a ^2$ where M is the max of $ f' $ on $[a, x]$
Quadratic approx.	$f(x) \approx P_{2,a}(x)$ See Table A	Better approximation than linear, at the cost of being harder to compute.
Binomial approx.	$(1 + g(x))^p \approx 1 + pg(X)$	Hypothesis: $g(x)$ is close to 0.
Right Riemann sum approx.	$\int_a^b f(x)dx \approx \sum_{i=1}^n f(x_i)\Delta x$ $\Delta x = \frac{b-a}{n}, \quad x_i = a + i\Delta x$	The bigger n is, the better the approximation, with error $ \text{error} \leq \frac{M}{2n}(b-a)^2$ where M is the max of $ f'(x) $ on $[a, b]$.

Table D: Basic Derivative Rules

Type	Function $f(x)$	Derivative $f'(x)$
Constant	c	0
Power	x^n (real n where defined)	nx^{n-1}
Constant multiple	$c f(x)$	$c f'(x)$
Sum / difference	$f(x) \pm g(x)$	$f'(x) \pm g'(x)$
Product rule	$f(x)g(x)$	$f'(x)g(x) + f(x)g'(x)$
Quotient rule	$\frac{f(x)}{g(x)}$	$\frac{f'(x)g(x) - f(x)g'(x)}{(g(x))^2}$
Chain rule	$f(g(x))$	$(f \circ g)'(x) = f'(g(x)) \cdot g'(x)$
Exponential	e^x	e^x
Ugly exponential	$a^x = e^{x \ln a}$ ($a > 0$)	$a^x \ln a$
Logarithm	$\ln x$ ($x > 0$)	$\frac{1}{x}$
Absolute value	$ x $ ($x \neq 0$)	$\frac{x}{ x }$
Integral function	$\int_a^t g(x)dx$	$g(t)$

Type	Function	Derivative	Indefinite integral
Constant	c	0	$\int c dx = cx + C$
Power	x^n ($n \neq -1$)	nx^{n-1}	$\int x^n dx = \frac{x^{n+1}}{n+1} + C$
Constant multiple	$c f(x)$	$c f'(x)$	$\int c f(x) dx = c \int f(x) dx$
Sum / difference	$f(x) \pm g(x)$	$f'(x) \pm g'(x)$	$\int (f \pm g) dx = \int f dx \pm \int g dx$
Product rule	$f(x)g(x)$	$f'(x)g(x) + f(x)g'(x)$	No general formula.
Quotient rule	$\frac{f(x)}{g(x)}$	$\frac{f'(x)g(x) - f(x)g'(x)}{[g(x)]^2}$	No general formula.
Chain rule, or Composition	$f(g(x))$	$(f \circ g)'(x) = f'(g(x))g'(x)$	Reverse chain rule Table B .
Exponential	e^x	e^x	$\int e^x dx = e^x + C$
Ugly exponential	a^x ($a > 0, a \neq 1$)	$a^x \ln a$	$\int a^x dx = \frac{a^x}{\ln a} + C$
Logarithm	$\ln x$	$\frac{1}{x}$	$\int \frac{1}{x} dx = \ln x + C$
Ugly logarithm	$\log_a x$ ($a > 0, a \neq 1$)	$\frac{1}{x \ln a}$	$\int \frac{1}{x \ln a} dx = \log_a x + C$

Table E: Basic Trigonometric Identities

Type	Identity
Pythagorean	$\sin^2 x + \cos^2 x = 1$
Pythagorean	$1 + \tan^2 x = \sec^2 x$
Pythagorean	$1 + \cot^2 x = \csc^2 x$
Quotient	$\tan x = \frac{\sin x}{\cos x}, \quad \cot x = \frac{\cos x}{\sin x}$
Even / odd	$\sin(-x) = -\sin x, \quad \cos(-x) = \cos x,$ $\tan(-x) = -\tan x$
Angle addition	$\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta$ $\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$
Double-angle	$\sin(2x) = 2 \sin x \cos x$ $\cos(2x) = \cos^2 x - \sin^2 x$

Trigonometric Derivative/Integrals

Function	Derivative	Notes
$\sin x$	$\cos x$	
$\cos x$	$-\sin x$	
$\tan x$	$\sec^2 x$	
$\cot x$	$-\csc^2 x$	
$\sec x$	$\sec x \tan x$	
$\csc x$	$-\csc x \cot x$	
$\arcsin x$	$\frac{1}{\sqrt{1-x^2}}$	Domain is $[0, 1]$
$\arccos x$	$-\frac{1}{\sqrt{1-x^2}}$	Domain is $[0, 1]$
$\arctan x$	$\frac{1}{1+x^2}$	

Integrand	Indefinite integral
$\tan x$	$\int \tan x \, dx = \ln \sec x + C$
$\sec x$	$\int \tan x \, dx = \ln \sec x + \tan x + C$
$\csc x$	$\int \csc x \, dx = -\ln \csc x + \cot x + C$

Table F: Things you need for curve sketching

Concept	How to detect
f increasing on an interval	$f'(x) > 0$ for all x in the interval.
f decreasing on an interval	$f'(x) < 0$ for all x in the interval.
Concave up on an interval	$f''(x) > 0$ for all x in the interval (or f' is increasing there).
Concave down on an interval	$f''(x) < 0$ for all x in the interval (or f' is decreasing there).
Critical point / critical number	A point c in the domain where $f'(c) = 0$ or $f'(c)$ does not exist.
Inflection point	A point c in the domain where $f''(c) = 0$.
Local maximum/minimum	See first derivative test in Table B